Real-Time DeepJSCC-Q on FPGA

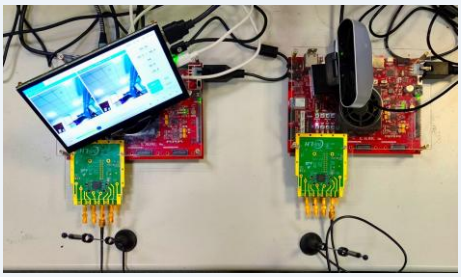
Wireless image transmission on two Zynq boards

30 fps
256 × 256 RGB
18,000 frames in 600.06 s

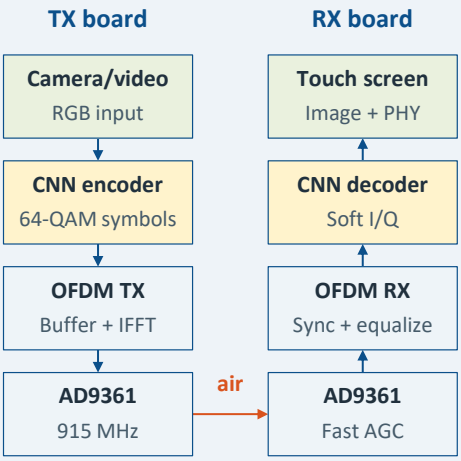
71.5 ms
PL frame latency
Frames overlap in the pipeline

+3.8 dB
High-SNR PSNR gain over SSCC
DIV2K 0802; equal symbol budget

Two-board wireless link



RX (left), TX (right).
2 × PYNQ-ZU + 2 × AD-FMCOMMS3.



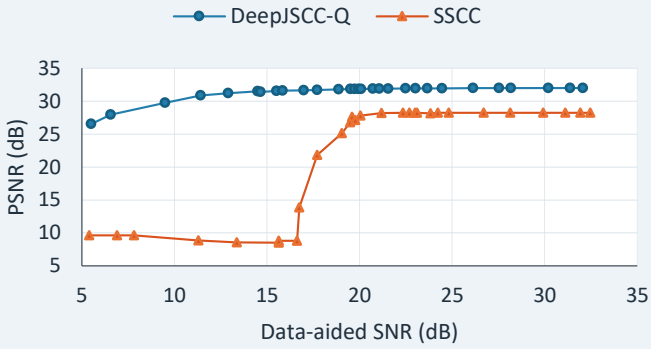
32,768 complex symbols/image
683 OFDM symbols/frame
20 MSPS PHY sample rate

Soft I/Q enters the decoder without hard decisions. Ethernet carries control and the reference preview.

PL latency excludes app/display.
Transfer includes buffers + RX PHY.

Measured quality over the air

DIV2K 0802; same PHY and 32,768 symbols/image.
SSCC: JPEG + convolutional code + 64-QAM.

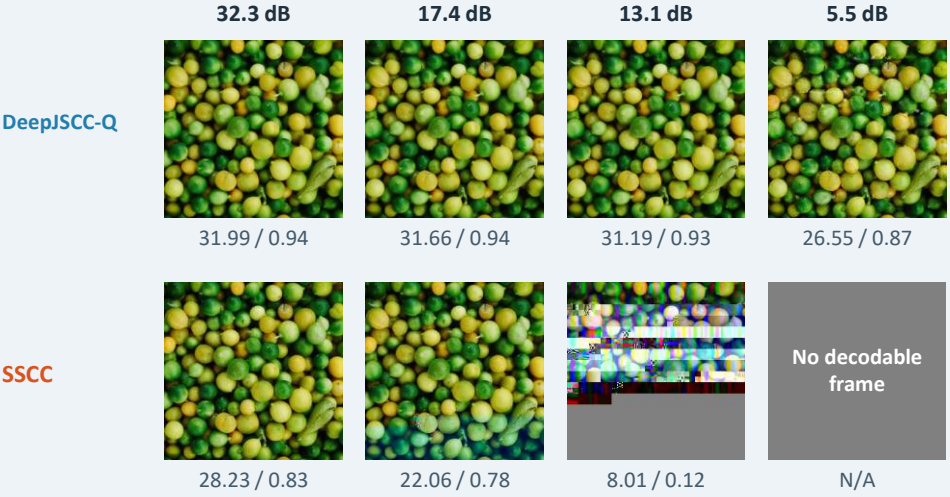


Original



256 × 256
RGB

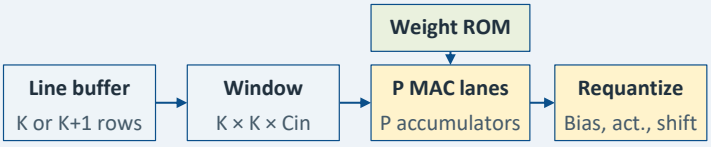
Reconstruction at four channel conditions



Matched TX attenuation; column SNR averages both schemes.
Curve: window means. Images: frame PSNR (dB) / SSIM.
Decode failures are scored as mid-gray.

Streaming FPGA implementation

250 MHz, W8A12, 2.33 GMAC/frame



Layer engines form a pipeline; P meets 30 fps.
On-chip buffers avoid full intermediate feature maps.

Routed TX / RX totals on xczu5eg

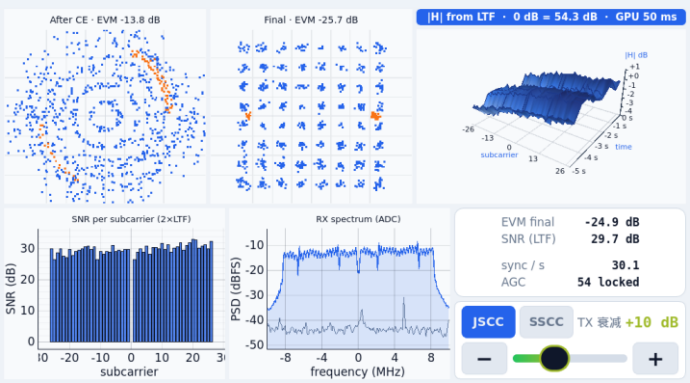
Design	LUT	BRAM36	URAM	DSP
TX	52,679	104.5	10	243
RX	70,768	123.5	25	394

Measured rails: TX 3.06 W / RX 3.71 W
PL + PS + DDR; excludes RF and input conversion losses.

PL frame latency: 71.5 ms



Live PHY diagnostics



DeepJSCC: TCCN 2019. DeepJSCC-Q: JSAIT 2022. Developed with Claude Code and Codex under human direction.